

best checkpoint, and evaluation is conducted on the test set using the clean labels. Further details of the training are provided in Appendix A.2.

Results. Figure 7 presents the results for both random feature models (left) and fully trained neural networks (right). In these experiments, we mixed synthetic and original data in the training set with varying coefficients, p_1 . As the proportion of synthetic data, p_2 , increases, the scaling laws slow down and eventually plateau. We observe a strong model collapse: only when p_2 approaches 0 does the collapse subside. The results are consistent across both cases, validating our theoretical predictions and demonstrating the applicability of our insights to more complex scenarios.

We also investigated how model size, specifically the hidden dimension of fully trained neural networks, affects model collapse. As shown in Figure 6, models with varying hidden dimensions were trained exclusively on the synthetic dataset with $p_2 = 1$. For training sets ranging from 10,000 to 50,000 samples, our results indicate that larger models are more susceptible to model collapse under the same validation and evaluation protocols. Notably, all these models remain in the interpolation regime, aligning with our theoretical predictions.

In summary, we find that the general trends observed in our asymptotic theory still hold: (1) there is significant model collapse, which only diminishes as the fraction of synthetic data approaches 0; (2) larger models exhibit a more severe model collapse.

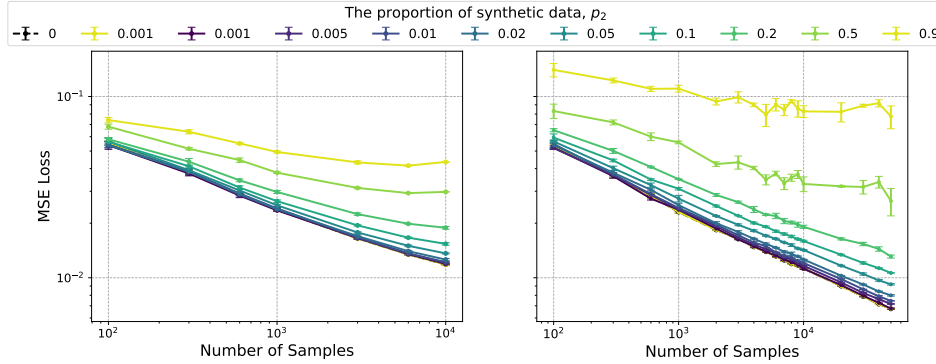


Figure 7: **Model collapse as a function of the proportion of synthetic data.** We use the MNIST dataset with regression loss. Error bars correspond to 5 runs. **Left**, Random feature model with hidden dimension 100,000. **Right**, Two-layer neural network of width (i.e hidden dimension) $m = 2000$.

4.2 Language Modeling

We then extend our focus to tasks beyond classification, specifically language modeling with GPT-2 models. The BabiStories dataset (Zhang et al., 2024a), a reproduction of TinyStories (Eldan & Li, 2023) using the Mixtral-8x7B open language model (Jiang et al., 2024) enables us to study language modeling with relatively small models in a compute-efficient and environmentally friendly way. It comprises stories generated by prompting large models to create narratives in simple language that a three-year-old child could understand, effectively simplifying the underlying language model.

Setup. We train a GPT-2-small model with 124 million parameters on the BabiStories dataset as the generator. Using the same story prompts, which include the story beginning and character names, the generator creates our synthetic dataset. We then mix this synthetic dataset with the original BabiStories, train, and evaluate model perplexity on a validation set derived from the original BabiStories. Detailed information on optimization and model configurations is provided in Appendix A.3.

Impact of Synthetic Data Proportion. We investigate the effect of varying the synthetic data proportion (p_2) on the model’s scaling in Figure 8 (left). Here, the x-axis represents the number of tokens used to train the model. In this experiment, the synthetic data is of high quality, as evidenced by the low training loss and coherent text generations, corresponding to the small c^2 (cf. Definition 1) case in our illustrative Figure 1. Consequently, even moderate amounts of synthetic data delay

the progression of the scaling laws, and we expect this to eventually lead to plateaus or at least very bad bad (i.e small) exponents in the final scaling laws as predicted in Dohmatob et al. (2024b) in the special case of training on synthetic data only.

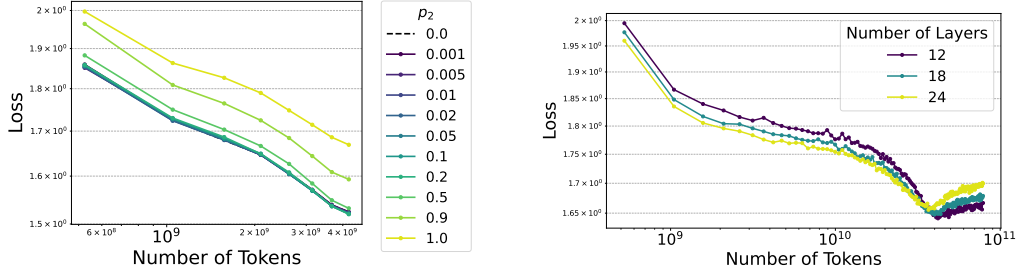


Figure 8: **Results on BabiStories with GPT-2 models.** Synthetic BabiStories is generated with a trained GPT-2-small with the same set of prompts. **(Left)** Impact of the proportion of synthetic data p_2 on model collapse in a language model with 12 layers. **(Right)** Impact of model size (number of layers) on model collapse. Here the model is trained on synthetic data only (i.e $p_2 = 1$). The loss is evaluated on the TinyStories test set.

Impact of Model Size. We next examine the impact of model size on training with synthetic data. In addition to the GPT-2-small model (12 layers), we introduce two larger models: one with 18 layers (166 million parameters) and another with 24 layers (204 million parameters). The embedding dimension and number of attention heads remain constant across all models. We generate a synthetic dataset 10 times larger than the original one to support the scaling of tokens. As shown in Figure 8 (right), larger (deeper) models maintain a lower test loss until the dataset size increases—likely exceeding the interpolation threshold—at which point smaller models begin to exhibit lower loss and reduced overfitting. This aligns with the predictions of Theorem 2 (also refer to Figure 1, 2, and the discussion just after the theorem), which suggest that larger models tend to amplify model collapse beyond the interpolation threshold. In Figure 8, we observe this amplification when the number of tokens exceeds 3×10^{10} . Conversely, the theory predicts that over-parameterized models help mitigate collapse, a trend we observe when the number of tokens is below 1×10^{10} , leading to improved performance of larger models.

5 Can Strategic Data Mixing Schemes Prevent Model Collapse?

Having established the occurrence of strong model collapse both theoretically and empirically, we now explore strategies to mitigate it and leverage synthetic data under stronger assumptions. We begin by assuming clear information about the data source and consider two data mixing methods: 1) weighted data mixing (Jain et al., 2024), 2) strategic iterative mixing, inspired by Ferbach et al. (2024).

5.1 Weighted Single-Step Data Mixing

For the purposes of studying scaling laws for learning on mixtures of real and surrogate data (e.g synthetic data), the setting considered in (Jain et al., 2024) consists in the following optimization problem:

$$\hat{w} = \arg \min_{w \in \mathbb{R}^n} \frac{(1 - \alpha)}{n_1} \sum_{(x_i, y_i) \in \mathcal{D}_1} (x_i^\top w - y_i)^2 + \frac{\alpha}{n_2} \sum_{(x_i, y_i) \in \mathcal{D}_2} (x_i^\top w - y_i)^2 + \lambda \|w\|^2. \quad (16)$$

This is an instance of weighted empirical risk minimization (Shimodaira, 2000; Vogel et al., 2021) where the weight the sample weight π_i is constant across each group: $\pi_i = (1 - \alpha)n/n_1 \simeq (1 - \alpha)/p_1$ for real samples vs $\pi_i = \alpha n/n_2 \simeq \alpha/p_2$ for synthetic samples. Thus $\alpha \in (0, 1)$ is a mixing coefficient for the two the two source of data; in particular $\alpha \rightarrow 0$ corresponds to only using real data for training, while $\alpha \rightarrow 1$ corresponds to only using surrogate data. Formula (16) replaces the formula for the weights vector $\hat{w}$ of the classical linear model f_{CL} (3).

For conciseness, as in Section 5 we focus on the isotropic case considered in Section 3.1 where the feature covariance matrices are $\Sigma_1 = \Sigma_2 = \Sigma = I_d$ and the shift matrix $\Delta := \text{cov}(w_1^* - w_2^*)$ has

the form $(c^2/d)I_d$ for some scalar $c > 0$. Further, let us consider the regime where $d/n \rightarrow 0$. In the language of our paper, one should think of this as corresponding to the proportionate scaling regime given in (4), and then letting $\phi \rightarrow 0^+$ (extremely under-parametrized regime). We have the following result.

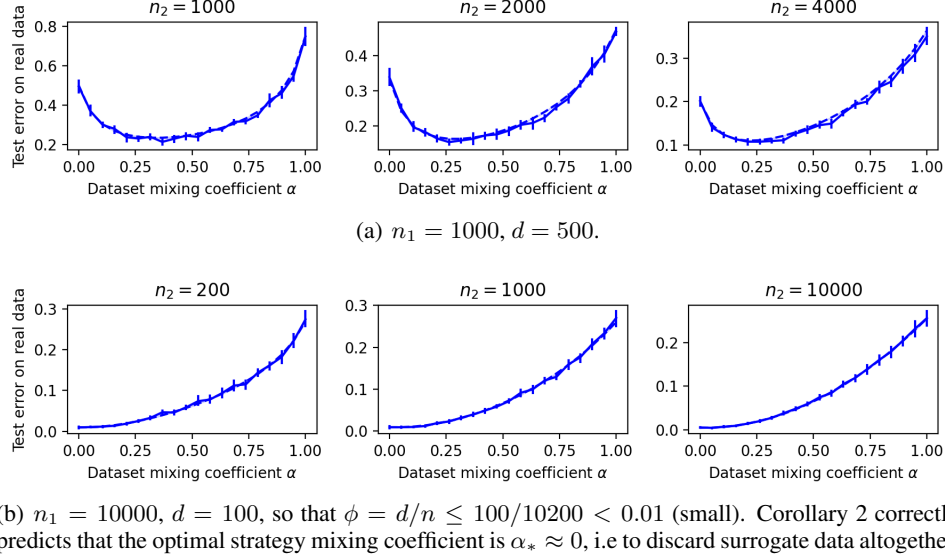


Figure 9: **Failure of naive real+surrogate data mixing to solve model collapse.** For this experiment, we use different several different values for the size of the real data n_1 and the synthetic data n_2 . Solid curves correspond to experiments while broken curves correspond to our theoretical prediction give in Corollary 2. Error-bars correspond to independent runs.

Corollary 2. Consider the proportionate scaling limit (4). For small $\phi = d/n$, it holds that

$$E_{test}(\hat{f}_{CL}) \simeq p_2^2 \alpha^2 c^2 + ((1 - \alpha)p_1 \sigma_1^2 + \alpha p_2 \sigma_2^2) \phi + O(\phi^2). \quad (17)$$

The formula given in (17) represents a U-shaped function of α , minimized when $\alpha = \alpha_*$, with

$$\alpha_* = \text{clip}_{[0,1]} \left(1 - \frac{p_1 \sigma_1^2 - p_2 \sigma_2^2}{2c^2} \phi \right). \quad (18)$$

Single-Step Mixing Does Not Avoid Model Collapse. It should be clear that if $\text{tr } \Delta = \Omega(1)$ and $\sigma_1, \sigma_2 = O(1)$, then $\alpha_* \rightarrow 0$; this corresponds to only using real data for training! In contrast any fixed value $\alpha \in (0, 1]$, leads a positive lower-bound on test error $E_{test}(\hat{f}_{CL}) \geq p_2^2 \alpha^2 c^2 \gtrsim c^2$; this is effectively model collapse. The situation is empirically confirmed in Figure 9.

5.2 Dynamic / Multi-Step Data Mixing

As in the case of single-step mixing discussed in Section 5.1, take for concreteness, take $\Sigma_1 = \Sigma_2 = \Sigma = I_d$ for the covariance matrices, and $\Delta = (c^2/d)\Sigma^{-1} = (c^2/d)I_d$. In this setup, the proposal of Ferbach et al. (2024) then becomes

- The quality parameter (cf. Def 1) of the synthetic data at the beginning (i.e at $t = 0$) is $c^2 = c_0^2$.
- At iteration $t + 1$, we mix $n_2 = p_2 n$ samples of synthetic data from a source having quality parameter $c^2 = c_t^2$, with $n_1 = n - n_2$ samples of real data to construct a penalized linear model $\hat{w}^{(t+1)}$ according to (3). This trained model generates the synthetic data with $c^2 = c_{t+1}^2$.

Thus, the idea is to iteratively enhance the quality of the synthetic data through bootstrapping.

Corollary 3. For large t , it holds in the limit (4) and then $\phi, \lambda \rightarrow 0^+$ that

$$E_{test}(\hat{f}_{CL}^{(t)}) \simeq \bar{E}/(1 - p_2^2) + \Theta(p_2^{2t}), \text{ where } \bar{E} \simeq \sigma^2 d/n, \quad \sigma^2 := p_1 \sigma_1^2 + p_2 \sigma_2^2. \quad (19)$$